

LAB REPORT #3 SIMULATION OF QUEUING SYSTEM USING PYTHON

OBJECTIVES

- o To simulate a single-server queuing system using Python.
- o To generate customer arrival and service times.
- o To calculate waiting time and completion time for each customer.
- o To visualize customer waiting time using a graph.
- o To understand the behaviour of a first-come first-served (FCFS) queue.

THEORY

A queuing system is a mathematical model used to represent situations where customers arrive for service, wait if the server is busy, receive service, and then leave the system.

Examples of queuing systems include:

- o Bank counters
- o Hospital registration desks
- o Ticket booking system
- o Call centers
- o CPU process scheduling

In this experiment, a single-server FCFS queue is simulated. Customers are served in the same order in which they arrive.

Important terms:

i) Arrival time (AT):

The time at which a customer enters the system.

ii) Service time (ST):

The time required to serve a customer.

iii) Waiting Time (WT):

The time a customer spends waiting in the queue before receiving service.

iv) Completion Time (CT):

The time at which a customer's service is completed.

Formulae used:

Waiting Time

Where:

- $(WT_i) =$ Waiting time of current customer.
- $(CT_{i-1}) =$ Completion time of previous customer.
- $(AT_i) =$ Arrival time of current customer.

The $\max(0, \dots)$ function ensures that waiting time never becomes negative.

Service start time

$$T_{start-i} = AT_i + WT_i$$

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Completion Time

Where:

- $(CT_i) =$ Completion Time
- $(ST_i) =$ Service time


```
import random
import matplotlib.pyplot as plt
```

```
n=10
```

```
arrival-time = [0]
```

```
service-time = []
```

```
waiting-time = [0]
```

```
completion-time = []
```

```
# Generate increasing arrival times
```

```
for i in range(1, n):
```

```
    arrival-time.append(arrival-time[i-1] + random.randint(1, 9))
```

```
# Generate service times
```

```
for i in range(n):
```

```
    service-time.append(random.randint(1, 5))
```

```
# First customer
```

```
completion-time.append(arrival-time[0] + service-time[0])
```

```
# Remaining customers
```

```
for i in range(1, n):
```

```
    waiting = max(0, completion-time[i-1] - arrival-time[i])
    waiting-time.append(waiting)
```

```
    start-service = arrival-time[i] + waiting
```

```
    completion-time.append(start-service + service-time[i])
```

```
# Display results
```

```
print("\n" * 60)
```

```
print("Customer":<10> "Arrival":<10> "Service":<10>
      "Waiting":<10> "Completion":<10>")
```

```
print("\n" * 60)
```

```
for i in range(n):  
    print(f"{i+1}: <10 } { arrival-time[i]:<10 } { service-time[i]:<10 }  
          { waiting-time[i]:<10 } { completion-time[i]:<10 }")  
print("-" * 60)
```

Plot waiting times

```
plt.plot(range(1, n+1), waiting-time, marker='o')  
plt.title("Waiting time for customers")  
plt.xlabel("Customer Number")  
plt.ylabel("Waiting Time")  
plt.grid(True)  
plt.show()
```


DISCUSSION

The Single Server Queue simulation using the first come first served (FCFS) scheduling algorithm was successfully implemented in Python.

The program generated random customer arrival and service times, calculated waiting and completion times, and displayed the results in both tabular and graphical formats.

The simulation illustrates the operation of a basic queuing system and shows waiting time is influenced by customer arrival patterns and service duration. It also highlights the importance of efficient service management in reducing customer waiting time and improving system performance. This experiment provides practical knowledge of queue simulation concepts and demonstrates the application of Python programming & data visualization in modeling real-world service systems.

CONCLUSION

Hence, The simulation of Queuing System using FCFS with Python Language has been successfully implemented.